

01 Multi-source Training Data

Vision Question Answer Data



Q: How many tigers are sitting on the bench?
A: Three.



Q: What is likely to happen next?
A: Move to the hanger.

General VQA

Embodied VQA

Egocentric Human Data



EgoDex



EgoVerse



VITRA

Robot Data



AGIBotWorld



DROID



RoboCOIN



InternData-A1



Bridge



Fractal

02 Unified World-Action Model

UniWAM

Instruction

Video frames

State

Tidy up the desk



VL Encoder

VAE Encoder

Action Encoder

Physical Reasoner

World Generator

Action Predictor

Joint Multimodal Attention

Language Head

Video Head

Action Head

Semantics

Subtask: Put the keys into drawer. Left arm: move forward 2cm. Right arm: tilt right 12 degrees, tilt forward 4 degrees"

Future frames

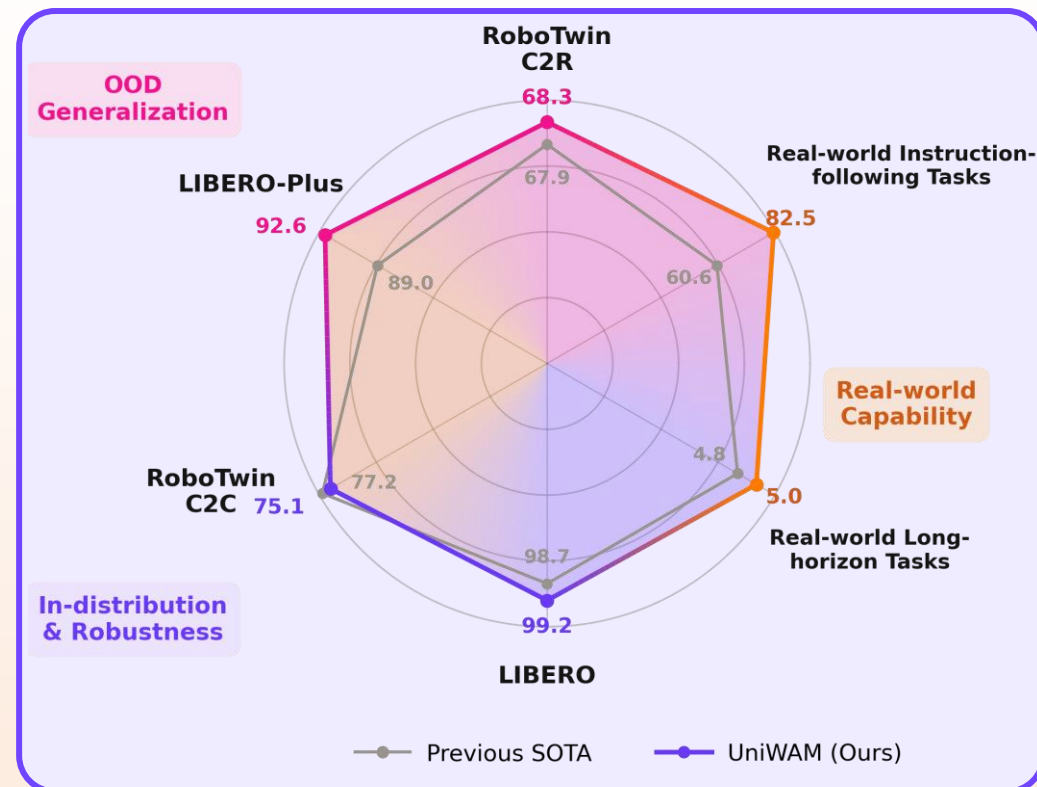


Action chunk

xyz: [0.12, 0.03, 0.25]
rpy: [0.00, 0.21, 0.07]
gripper: [1.0]

03 Benchmark Performance

Evaluations across 6 key simulation and real-world settings



Results & Takeaways

1 Strong In-distribution Performance

State-of-the-art results on LIBERO and RoboTwin C2C

2 Competitive OOD Generalization

Consistent performance across LIBERO-Plus and RoboTwin C2R

3 Promising Real-world capability

Strong results on Real-world Instruction-following Tasks and Long-horizon tasks